

RIGID TRANSFORMATION

Computer Vision with Hüseyin Özdemir

Rigid transformation is a combination
of rotation and translation

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of rotation and translation

Rigid means the transformation
preserves shape and scale

Rotation and translation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

θ is clockwise
rotation angle

Rotation and translation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

θ is clockwise rotation angle

The diagram illustrates the transformation equation with four labels and arrows pointing to the corresponding parts of the equation:

- Output coordinate vector**: Points to the vector $\begin{bmatrix} x' \\ y' \end{bmatrix}$ (blue box).
- Rotation matrix**: Points to the matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (red box).
- Input coordinate vector**: Points to the vector $\begin{bmatrix} x \\ y \end{bmatrix}$ (blue box).
- Translation vector**: Points to the vector $\begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$ (red box).

Rotation and translation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

θ is clockwise
rotation angle

Rotation and translation in homogeneous coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & b_0 \\ \sin \theta & \cos \theta & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & b_0 \\ \sin \theta & \cos \theta & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

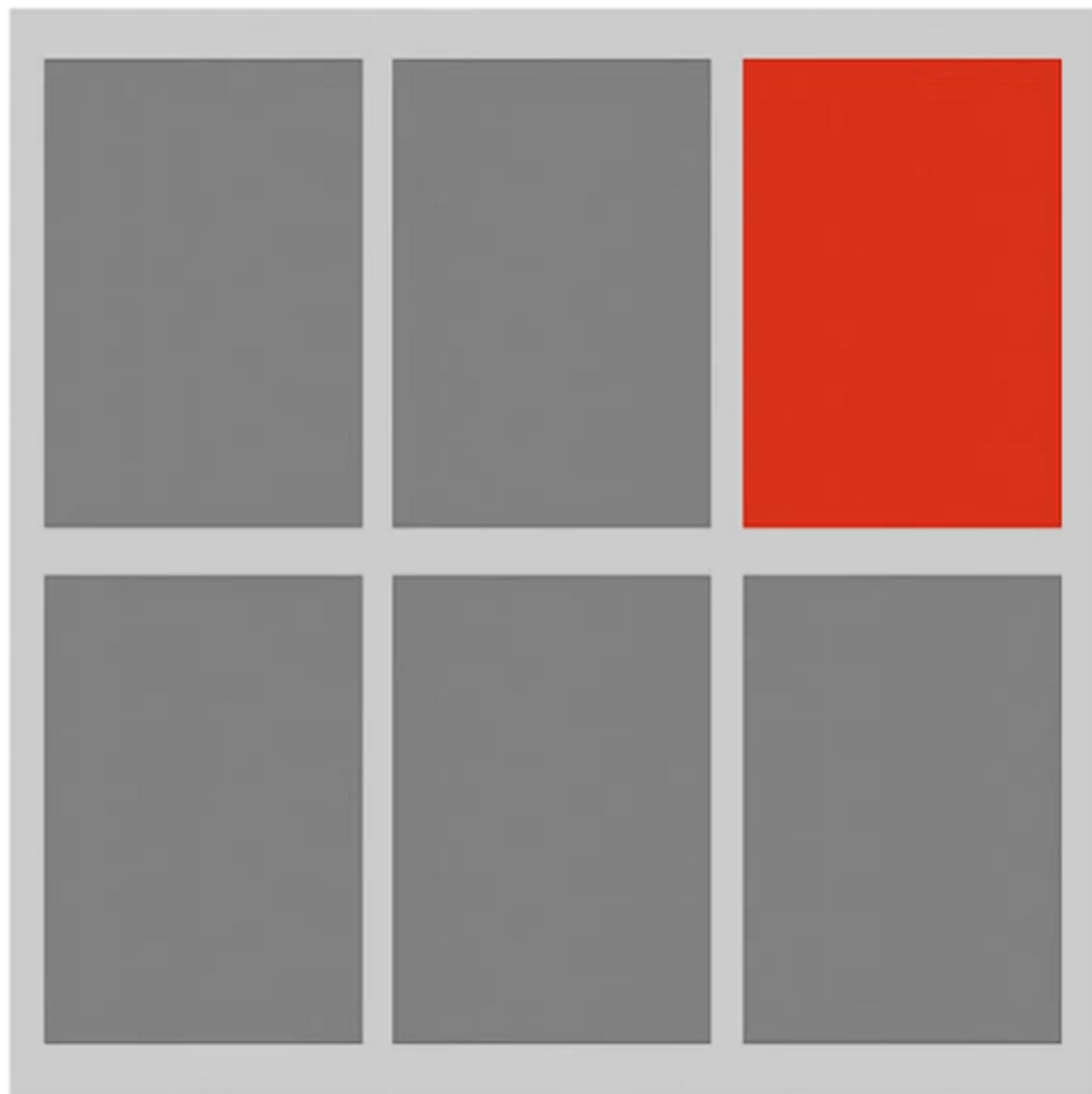
Rigid transformation is controlled with 3 parameters

3 Degrees of Freedom (DoF)

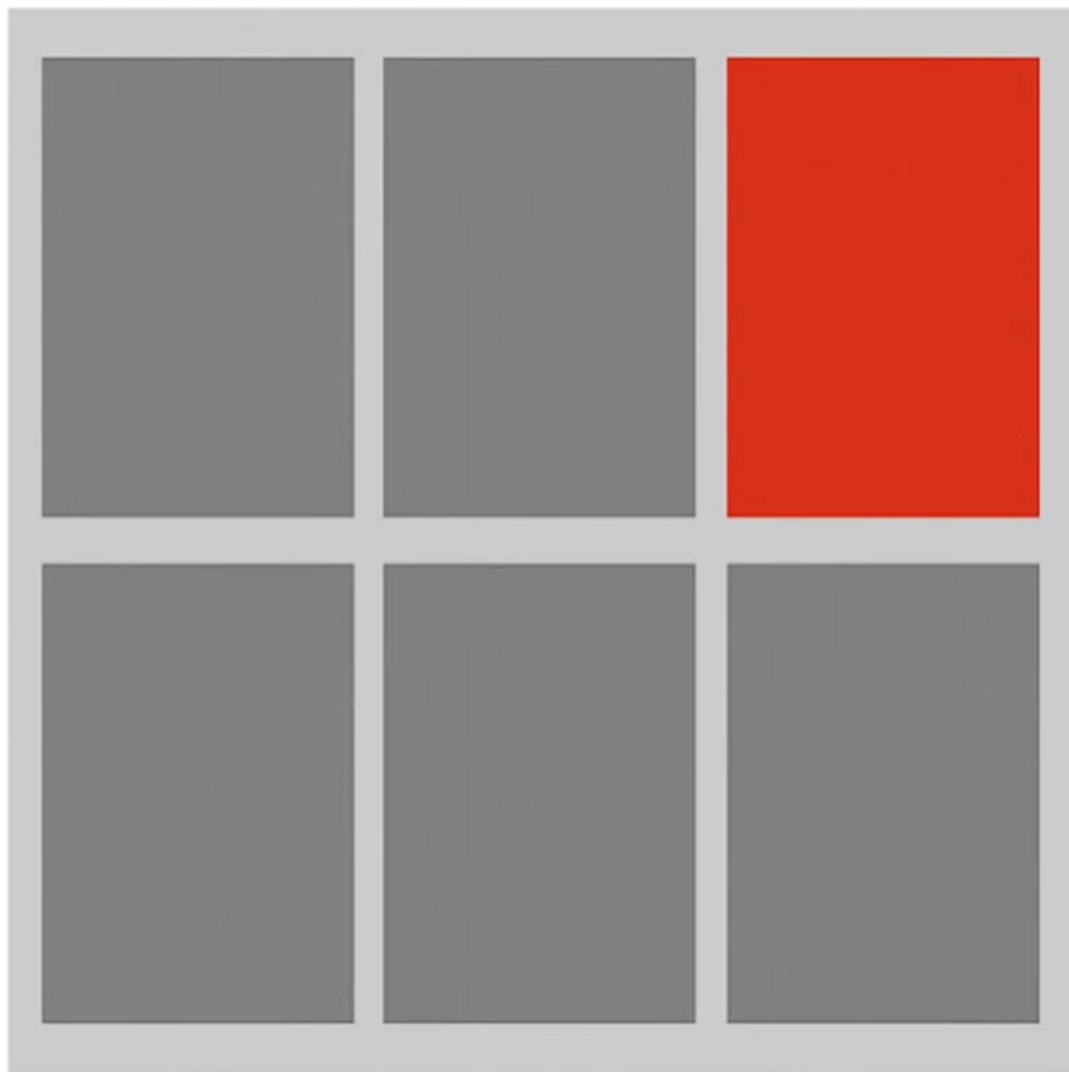
θ , b_0 and b_1

Rigid Transformation preserves lines,
lengths and angles

Parallel lines remain parallel

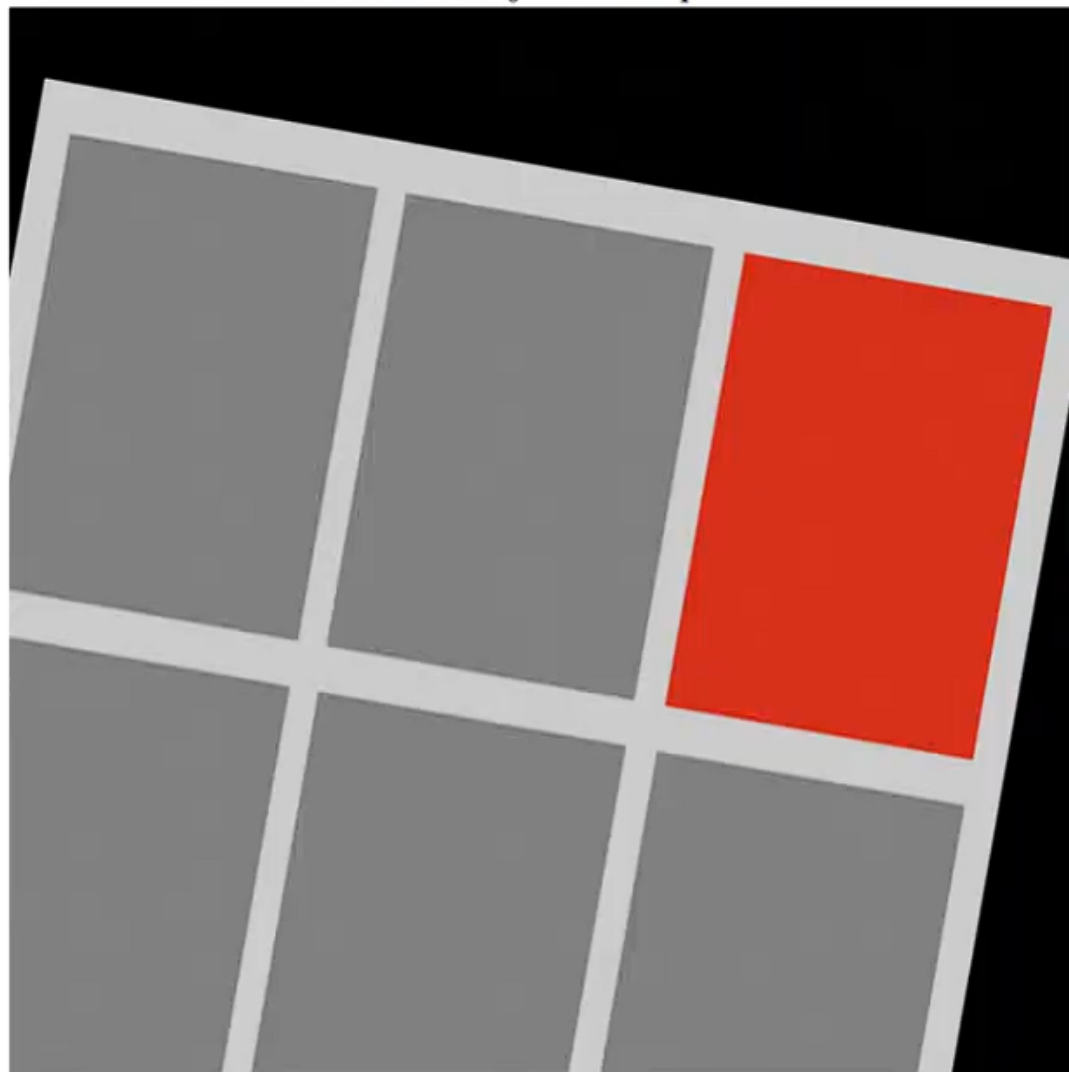


Original Image



Original Image

$$\theta = 10^\circ, b_0 = 30, b_1 = 60$$



Rotated and Translated Image

Reference

- <https://www.youtube.com/watch?v=sQiBGftz -c>